

December 2025 reconstruction audit

Supplement to Appendix C of *Public Trader Identity: Adverse Selection and Return Predictability* (Zhai, 2026). Score Dec. 1–10, fit Dec. 11–20, evaluation Dec. 21–31; BTC, ETH, SOL from the public Zenodo release (DOI 10.5281/zenodo.18184441).

Table 1: December 2025 reconstruction audit: timing anchors and closure intervals

Coin	Input rows	Source exact	Trusted	Demoted	Stale rows	Stale seg.	Closures	Closed (s)	Retained (%)	Viol.
BTC	1.700B	1.697B	1.697B	1.9k	2.94M	2,841	8,122	1,160.4	99.9567	0
ETH	1.082B	1.080B	1.080B	2.8k	2.10M	2,661	2,885	2,372.2	99.9114	0
SOL	582.74M	581.61M	581.61M	1.7k	995.4k	1,341	2,069	324.4	99.9879	0

Available time is the authoritative lifecycle-sequence clock. Exact LNDS repair demotes only anchors incompatible with global order; closure intervals suppress uncertain state. Final checks require zero unresolved diffs at gate reopen, zero available-time and equal-time gate-order violations, and zero crossed valid BBO rows. Reconstruction covers score Dec. 1–10, fit Dec. 11–20, and evaluation Dec. 21–31.

Table 2: December 2025 reconstruction audit: replay and independent cross-checks

Panel A: reset-gated replay volume

Coin	Groups	Valid groups	Resets	Reset rows	Warm-up rows	Discarded live
BTC	27.74M	27.18M	1,002	288.4k	64.63M	2.95M
ETH	26.21M	25.85M	616	146.3k	31.91M	1.56M
SOL	25.32M	25.10M	374	41.5k	9.91M	502.9k

Panel B: replay integrity and gate exclusion

Coin	Missing remove	Missing update	Total excluded (s)	Crossed valid
BTC	2.89M	353.3k	55,971	0
ETH	1.56M	235.4k	38,156	0
SOL	499.7k	79.1k	22,063	0

Panel C: trade-location validation

Coin	Fills	Matched (%)	Direction (%)	At best (%)	Age p95 (ms)	Pre-touch (%)
BTC	11.92M	100.000	86.52	45.34	170.2	97.76
ETH	6.84M	100.000	89.81	62.18	175.5	96.51
SOL	3.10M	100.000	92.06	68.44	185.7	97.75

Panel D: status-book validation

Coin	Status cuts	Price exact (%)	Mid p95 (bp)	Best depth p95 (%)	Near depth p95 (%)
BTC	2.60M	99.782	0.000	99.91	25.80
ETH	2.62M	99.688	0.000	92.07	22.44
SOL	2.63M	99.824	0.000	41.21	11.13

Panel E: reset-age depth stability

Coin	Min. reset age (s)	Cuts	Gates	Price exact (%)	Mid p95 (bp)	Best depth p95 (%)	Near depth p95 (%)
BTC	1.0	71	69	88.732	0.410	422.01	487.65
BTC	5.0	329	68	96.049	0.000	814.27	91.12
BTC	10.0	1.3k	65	98.963	0.000	209.85	51.31
BTC	30.0	1.8k	61	99.044	0.000	201.28	54.50
BTC	60.0	79.0k	59	99.728	0.000	97.47	18.55
ETH	1.0	17	17	100.000	0.000	515.75	158.86
ETH	5.0	89	18	98.876	0.000	62.01	58.95
ETH	10.0	346	18	98.266	0.000	137.55	44.82
ETH	30.0	465	16	98.495	0.000	102.59	17.55
ETH	60.0	81.8k	16	99.678	0.000	51.51	5.13
SOL	1.0	16	16	93.750	0.000	89.75	286.43
SOL	5.0	86	17	93.023	0.395	99.50	77.95
SOL	10.0	339	17	96.165	0.000	54.17	97.94
SOL	30.0	508	17	98.228	0.000	99.61	84.56
SOL	60.0	81.8k	18	99.774	0.000	26.85	3.30

Total excluded seconds combine timing closures and configured initial and post-reset warm-up. Missing remove/update counts are lifecycle messages whose order ID is absent from the live-order map; replay counts and ignores the message rather than creating an order. Book validity is governed separately by timing gates, warm-up, and independent cross-checks. Resets discard the live book before a new gate reopens. Trade checks use the independent maker-fill ledger; status checks use the independently reconstructed status-only book. Panel E reports the selected 1, 5, 10, 30, and 60-second post-reset thresholds. Final hard-gate parameters are initial warm-up 3600s, post-reset warm-up 60s, and maximum quote age 120s. Reconstruction covers score Dec. 1–10, fit Dec. 11–20, and evaluation Dec. 21–31.

Table 3: December 2025 timing-qualified Stage-0 data volume

Panel A: reconstruction and gate admission

Coin	Timed diffs	Book-valid BBO	Timing excluded	Admitted BBO	Resets	Gates
BTC	1.697B	27.18M	7.9k	27.17M	1,002	8,699
ETH	1.080B	25.85M	2.9k	25.85M	616	3,360
SOL	581.75M	25.10M	2.0k	25.10M	374	2,335

Panel B: admitted ledgers and events

Coin	Checkpoint rows	Fill rows	Maker fills	Events v2	Event notional (\$bn)
BTC	117.8k	23.84M	11.14M	4.28M	75.06
ETH	95.4k	13.69M	6.47M	3.62M	42.57
SOL	47.4k	6.20M	3.00M	1.62M	9.84

Panel C: locked research windows

Window	Qualifying aggressive events	Event notional (\$bn)
Score: Dec. 1–10	3.96M	51.27
Fit: Dec. 11–20	3.23M	43.57
Evaluation: Dec. 21–31	2.34M	32.62

Book-valid BBO is the mechanical replay-valid count. Admitted BBO additionally excludes every declared timing-uncertainty closure and is certified to have exact own-gate coverage. Counts are produced only after the final timing, replay, trade/status reconciliation, and Stage-0 contract checks pass. An independently hashed contract covers all 93 daily checkpoints and 90 cross-day boundaries. TWAP and flagged liquidations are excluded from qualifying events; liquidation-status observability is reported separately in the Part-1 sensitivity. The reported windows are score Dec. 1–10, fit Dec. 11–20, and evaluation Dec. 21–31.